

A disproof of conjecture 00000000750: the exact eventual period of the reduced Mahler coefficient sequence is not $p^{k-1}(p-1)$

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Abstract

We refute conjecture 00000000750. The conjecture asserts that for the mod p^k reduction of a 1-Lipschitz map on $\mathbb{Z}_p$, the Mahler coefficient sequence $(a_n \bmod p^k)$ is eventually periodic with *exact* period $p^{k-1}(p-1)$, “which cannot be shortened”. We exhibit the map $f(x) = x + 1$, which is an isometry of $\mathbb{Z}_p$ (hence 1-Lipschitz), is Haar-measure-preserving, and is minimal. Its Mahler coefficient sequence is

$$(a_n)_{n \geq 0} = (1, 1, 0, 0, 0, \dots).$$

Reduced modulo p^k this sequence is eventually constant 0, so its minimal eventual period is 1. For the test pairs $(p, k) \in \{(2, 2), (2, 3), (3, 1), (3, 2), (5, 1), (5, 2)\}$ the conjectured value $p^{k-1}(p-1)$ is respectively 2, 4, 2, 6, 4, 20, never 1. Hence the conjecture is false. We also record that the conjecture’s quotation of Anashin’s criterion appears to be misstated: the correct divisibility condition is $p^{\nu_p(n!)} \mid a_n$, not $p^n \mid a_n$; the quoted form already fails for $f(x) = x + 1$ because $p \nmid a_1 = 1$ whenever $p > 1$.

1 Setup: Mahler coefficients

Let p be a prime and let $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ be continuous. Every such function has a unique Mahler expansion

$$f(x) = \sum_{n \geq 0} a_n \binom{x}{n}, \quad a_n = (\Delta^n f)(0) = \sum_{i=0}^n (-1)^{n-i} \binom{n}{i} f(i), \quad (1)$$

where Δ is the forward difference operator

$$(\Delta f)(x) = f(x+1) - f(x), \quad (\Delta^{n+1} f) = \Delta(\Delta^n f).$$

The numbers $a_n \in \mathbb{Z}_p$ are the *Mahler coefficients* of f .

Remark 1 (Anashin’s criterion). A standard form of Anashin’s criterion states that a continuous $f : \mathbb{Z}_p \rightarrow \mathbb{Z}_p$ is 1-Lipschitz (equivalently, the coefficients satisfy the compatibility needed for the expansion to converge) precisely when

$$p^{\nu_p(n!)} \mid a_n \quad \text{for all } n \geq 0, \quad (2)$$

where $\nu_p(n!) = \sum_{j \geq 1} \lfloor n/p^j \rfloor$. The conjecture under consideration quotes this condition as “ $p^n \mid a_n$ ”. That quoted form is not equivalent to (2): for $f(x) = x + 1$ one has $a_1 = 1$, so $p \mid a_1$ fails for every $p > 1$, even though f is an isometry and certainly 1-Lipschitz. This is a supporting observation about the wording of the conjecture; the refutation below does not rely on it.

2 The counterexample $f(x) = x + 1$

Work with $f(x) = x + 1$ on $\mathbb{Z}_p$.

Lemma 2. *The map $f(x) = x + 1$ has the following properties.*

1. *It is an isometry of $\mathbb{Z}_p$: $|f(x) - f(y)|_p = |x - y|_p$ for all $x, y \in \mathbb{Z}_p$. In particular it is 1-Lipschitz.*
2. *It preserves Haar measure (translation is measure-preserving).*
3. *It is minimal: the orbit $\{f^{\circ n}(0) : n \geq 0\} = \mathbb{N} = \{0, 1, 2, \dots\}$ is dense in $\mathbb{Z}_p$.*

Proof. (1) $f(x) - f(y) = x - y$, so the p -adic absolute values agree. (2) f is a translation, and Haar measure on the locally compact group $\mathbb{Z}_p$ is translation invariant. (3) $f^{\circ n}(0) = n$, and the natural numbers are dense in $\mathbb{Z}_p$ (the residues mod p^m exhaust every ball). $\square$

Proposition 3 (Mahler coefficients). *For $f(x) = x + 1$ the Mahler coefficient sequence is*

$$(a_0, a_1, a_2, a_3, \dots) = (1, 1, 0, 0, 0, \dots).$$

Equivalently, $(\Delta^n f)(0) = 0$ for every $n \geq 2$.

Proof. First, $\Delta(x + 1) = 1$ is the constant function 1:

$$(\Delta f)(x) = f(x + 1) - f(x) = (x + 2) - (x + 1) = 1.$$

Hence $\Delta^n(x + 1) = 0$ for every $n \geq 1$ applied to this constant, i.e. $\Delta^2 f = \Delta 1 = 0$ and all higher differences vanish. Evaluating at $x = 0$:

$$a_0 = f(0) = 1, \quad a_1 = (\Delta f)(0) = 1, \quad a_n = (\Delta^n f)(0) = 0 \quad (n \geq 2).$$

Equivalently, the Mahler expansion of f terminates: $x + 1 = \binom{x}{0} + \binom{x}{1}$, matching $a_0 = a_1 = 1$ and $a_n = 0$ for $n \geq 2$. $\square$

3 Periodicity modulo p^k

Fix a prime p and an integer $k \geq 1$, and reduce the sequence (a_n) modulo p^k .

Theorem 4. *The reduced sequence $(a_n \bmod p^k)_{n \geq 0}$ is*

$$(1, 1, 0, 0, 0, \dots) \bmod p^k,$$

which is eventually constant (equal to 0). Consequently its minimal eventual period is

$$T_{\text{actual}} = 1.$$

Proof. Since $p^k \geq 2$, we have $1 \bmod p^k = 1 \neq 0$, so the first two terms remain 1, 1. By Proposition 3, $a_n = 0$ for all $n \geq 2$, so $a_n \bmod p^k = 0$ for all $n \geq 2$; the tail is constant, and the smallest positive eventual period of a constant tail is 1. $\square$

Theorem 5 (Refutation of conjecture 00000000750). *The conjectured exact period is*

$$T_{\text{claimed}} = p^{k-1}(p-1).$$

For the test pairs

$$(p, k) \in \{(2, 2), (2, 3), (3, 1), (3, 2), (5, 1), (5, 2)\}$$

the claimed value equals

$$2, 4, 2, 6, 4, 20$$

respectively, whereas the actual minimal eventual period is 1 in every case. More generally, for $p > 1$ and $k \geq 1$ one has $p^{k-1}(p-1) = 1$ only for $(p, k) = (2, 1)$, so for every pair with $(p, k) \neq (2, 1)$ the claimed period differs from the actual period 1. Therefore the assertion that the exact period is $p^{k-1}(p-1)$ and “cannot be shortened” fails.

Proof. By Theorem 4 the actual period is 1. The listed values of $p^{k-1}(p-1)$ are obtained by direct computation: $2^1 \cdot 1 = 2$, $2^2 \cdot 1 = 4$, $3^0 \cdot 2 = 2$, $3^1 \cdot 2 = 6$, $5^0 \cdot 4 = 4$, $5^1 \cdot 4 = 20$. None equals 1. For the general claim: $p^{k-1} \geq 1$ and $p-1 \geq 1$ for $p > 1$, so $p^{k-1}(p-1) \geq p-1 \geq 1$; if $p \geq 3$ then $p-1 \geq 2$, and if $p = 2$ then $p-1 = 1$ and $p^{k-1}(p-1) = 2^{k-1}$, which equals 1 exactly when $k-1 = 0$, i.e. $k = 1$. Thus the unique solution of $p^{k-1}(p-1) = 1$ with $p > 1$, $k \geq 1$ is $(2, 1)$. $\square$

Remark 6. The hypotheses of the conjecture are satisfied by the counterexample: by Lemma 2, $f(x) = x+1$ is 1-Lipschitz, Haar-measure preserving, and minimal. The conjecture asserts that the exact period is attained “simultaneously with the measure-preserving property of minimal maps”; f is such a map, yet its coefficient sequence has eventual period 1. Hence the universal claim is false.

Verification summary

- `lean4/Main.lean` formalises the finite-difference core in core Lean (only `Int/Nat`, `import Std`, no `Mathlib`): the first difference of $x+1$ is the constant 1, the second and all higher differences vanish, hence the Mahler coefficients are $(1, 1, 0, 0, \dots)$; it also proves by `decide` that $p^{k-1}(p-1) \neq 1$ for the listed pairs.
- `reproduce.py` computes the Mahler coefficients by finite differences and compares the exact eventual period with the claimed value for the listed (p, k) ; it prints `PASS`.
- `lean4/Check.lean` prints the axiom dependencies of each theorem.